

JITEN BHALAVAT

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EDUCATION

University of Maryland, College Park

Aug 2024 - Expected May 2026

Master of Science in Applied Machine Learning

College Park, Maryland

Coursework: Machine Learning, Deep Learning, Natural Language Processing, Advanced Optimization, Cloud Computing

Charotar University of Science and Technology (CHARUSAT)

Sept 2020 - May 2024

Bachelor of Technology in Information Technology

Gujarat, India

Coursework: Data Structures, Full-Stack Development, Database Management System, Network Security

SKILLS

Languages and Databases: Python, C/C++, Java, JavaScript, SQL, HTML, Node.js, React, PostgreSQL, MySQL, MongoDB

Frameworks and Libraries: PyTorch, TensorFlow, Keras, HuggingFace, Scikit-learn, Flask, FastAPI, Transformers

Generative AI: LLMs, AI Agents, Vector Databases, Prompt Engineering, Fine-tuning, LangGraph, CrewAI, Autogen.

Tools & DevOps: Git, GitHub, Docker, Kubernetes, AWS, GCP, Azure, Tableau, PowerBI, MLFlow

EXPERIENCE

Machine Learning Engineer Intern

Sept 2023 - Apr 2024

Plutomen Technologies Pvt. Ltd.

Gujarat, India

- Architected and delivered a scalable production **RAG pipeline** (LlamaIndex + LangChain + Qdrant) serving real-time inference at i200ms latency; designed **end-to-end** from data ingestion to Flask REST API deployment with PostgreSQL
- Drove data extraction accuracy from **45% to 89%** and eliminated 15 hrs/week of manual review — collaborated closely with product and QA teams in an Agile SDLC to define acceptance criteria and ship incrementally.
- Engineered anomaly detection using Statistical analysis, Isolation Forest, and DBSCAN; instrumented observability monitoring that reduced false positives by 25%, ensuring reliable and accountable production inference.
- Improved chatbot answer relevance by 30% through iterative A/B testing of chunking strategies and semantic search configurations across Qdrant and Pinecone vector stores.

Research Assistant

Apr 2023 - Jun 2023

Charotar University of Science and Technology

Gujarat, India

- Trained a U-Net CNN architecture for MRI image segmentation, achieving **91% accuracy**
- Evaluated model performance using IoU and accuracy metrics, outperforming baseline CNNs by **15%**

Machine Learning Engineer Intern

May 2022 - Jun 2022

NXON Pvt. Ltd.

Gujarat, India

- Developed transformer-based **AI solution** using Parameter-Efficient Fine-Tuning (**PEFT**) with **LoRA** on T5, reducing trainable parameters by **99%** while achieving 60% faster training convergence on **distributed multi-GPU infrastructure**
- Built scalable data preprocessing and training pipeline with PyTorch DDP across 60K+ training steps, implementing automated checkpointing, experiment logging, and model evaluation workflows that reduced debugging cycles by 40%
- Optimized model performance across 10 programming languages using **CodeBLEU** evaluation metrics and **beam search**, enabling robust multi-language code intelligence

PROJECTS

ClassTopper: Full Stack Multi-modal RAG Platform

- Led 3-person team building multi-modal RAG chatbot for videos and documents, improving study efficiency by 40%
- Enhanced content retention by **60%** via mind-mapping feature, visualizing topic relationships for better knowledge recall
- Engineered adaptive quiz system generating personalized assessments from knowledge gap analysis, boosting test performance by 27% and identifying weak areas with 85% accuracy

Cloud LLM Inference Benchmark

- Applied responsible evaluation practices, tracked throughput, latency percentiles, and GPU memory across runs using structured logging for reproducible benchmarking results
- Deployed **GPU-accelerated inference** on AWS EC2 (NVIDIA A10G) with **Docker and CUDA**
- Benchmarked **vLLM vs SGLang** frameworks for GPU memory and concurrency optimization

Expense Tracker using MCP

- Deployed MCP server on FastMCP Cloud, prototyping agent-based workflow for finances through multi-agent tooling
- Integrated MCP Client as an AI-assisted workflow, allowing users to query, and analyze expenses via conversational AI
- Leveraged a lightweight **SQLite backend** for fast, local, and secure storage with instant retrieval and real-time insights.

CERTIFICATIONS AND ACHIEVEMENTS

Google Cloud Computing - Google Cloud Educational Badges (9 Badges)

AWS Academy Graduate - AWS Academy Machine Learning Foundations and AWS Academy Cloud Developing